

# Covariance/Variance Decomposition

Here we show that the population covariance can be expressed as a difference equation.

For  $\Delta_k a_t \equiv a_{t+k} - a_t$  (the difference) and  $\bar{a} \equiv \frac{1}{T} \sum_{t=1}^T a_t$  (the mean)

Given sequences  $\{a_t\}_{t=1}^T$  and  $\{b_t\}_{t=1}^T$ , we have

$$\frac{1}{T} \sum_{t=1}^T (a_t - \bar{a})(b_t - \bar{b}) = \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} \Delta_k a_t \Delta_k b_t$$

## Strategy

The population covariance can be written as follows:

$$\frac{1}{T} \sum_{t=1}^T (a_t - \bar{a})(b_t - \bar{b}) = \frac{1}{T} \sum_{t=1}^T (a_t b_t - \bar{a} b_t - a_t \bar{b} + \bar{a} \bar{b}) \quad (1)$$

$$= \left( \frac{1}{T} \sum_{t=1}^T a_t b_t \right) - \bar{a} \bar{b} \quad (2)$$

so the proof strategy is to show that the difference equation can be reduced to the same form.

## Proof

Since  $\Delta_k a_t \equiv a_{t+k} - a_t$  we can expand the difference equation as:

$$\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} \Delta_k a_t \Delta_k b_t = \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} (a_{t+k} - a_t) (b_{t+k} - b_t) \quad (3)$$

$$= \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} (a_{t+k} b_{t+k} + a_t b_t) \quad (4)$$

$$- \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_t b_{t+k} - \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_{t+k} b_t \quad (5)$$

and we'll evaluate each of the 4 terms in turn.

**term 1:**

Making a change of variable:  $s = t + k$  and noting that when  $t$  ranges from 1 to  $T - k$ ,  $s$  ranges from  $1 + k$  to  $T$ , we have:

$$\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_{t+k} b_{t+k} = \frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{s=k+1}^T a_s b_s \quad (6)$$

$$= \frac{1}{T^2} \sum_{s=2}^T \sum_{k=1}^{s-1} a_s b_s \quad (7)$$

$$= \frac{1}{T^2} \sum_{s=2}^T (s-1) a_s b_s \quad (8)$$

where in the second-last line we note that  $a_s b_s$  doesn't depend on  $k$  so we count each  $a_s b_s$  exactly  $s - 1$  times for  $s \in [2, T]$  (i.e. due to the inner sum). Note that this is

$$\frac{1}{T^2} (a_2 b_2 + 2a_3 b_3 + 3a_4 b_4 + \dots + (T-2) a_{T-1} b_{T-1} + (T-1) a_T b_T)$$

**term 2:**

The second term is

$$\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_t b_t$$

and it can be decomposed directly to:

$$\frac{1}{T^2} ((T-1)a_1 b_1 + (T-2)a_2 b_2 + (T-3)a_3 b_3 + \dots + a_{T-1} b_{T-1})$$

i.e.  $\frac{1}{T^2}$  times the following sum of  $T - 1$  values:

$$\begin{aligned} &+ (a_1 b_1 + a_2 b_2 + \dots + a_{T-3} b_{T-3} + a_{T-2} b_{T-2} + a_{T-1} b_{T-1}) \\ &+ (a_1 b_1 + a_2 b_2 + \dots + a_{T-3} b_{T-3} + a_{T-2} b_{T-2}) \\ &\quad \vdots \\ &+ (a_1 b_1 + a_2 b_2) \\ &+ (a_1 b_1) \end{aligned}$$

So combining the first and second terms we get  $\frac{T-1}{T^2} \sum_{t=1}^T a_t b_t$ , and we're on our way towards showing that

$$\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} \Delta_k a_t \Delta_k b_t = \left( \frac{1}{T} \sum_{t=1}^T a_t b_t \right) - \bar{a} \bar{b}$$

### term 3:

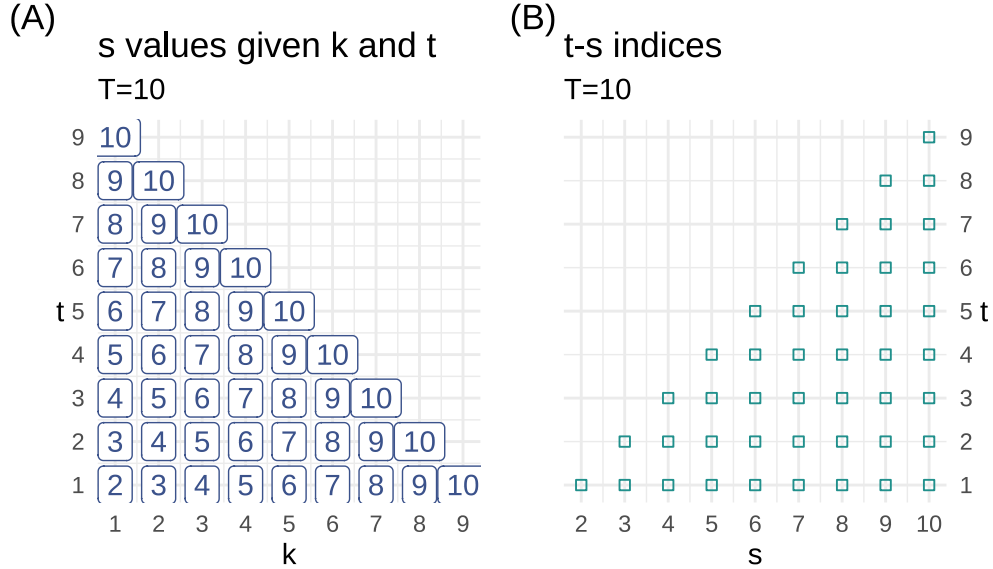
The third term is  $-\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_{t+k} b_t$  and again we will use the change of variables  $s = t + k$ .

Before changing variables, the outer sum goes from  $k = 1$  to  $k = T - 1$  and the inner sum goes from  $t = 1$  to  $t = T - k$ . The  $(k, t)$  indices used in  $-\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_{t+k} b_t$  are shown in figure (A) below (for  $T = 10$ ), along with the  $s = t + k$  values.

With the change of variable, the outer sum goes from  $s = t + k = 2$  to  $s = t + k = T$  and the inner sum goes from  $t = 1$  to  $t = s - 1$ . The  $(s, t)$  indices used to sum  $-\frac{1}{T^2} \sum_{s=2}^T \sum_{t=1}^{s-1} a_s b_t$  are shown in figure (B) below (for  $T = 10$ ).

The first index ( $s$ ) is along the horizontal axis, while the second index ( $t$ ) is along the vertical axis.

```
T <- 10
res <- 1:(T-1) |> # outer sum
  purrr::map(
    \(k){ 1:(T-k) |> # inner sum
      purrr::map(\(t)data.frame("k"=k, "t"=t, "s"=t+k)) }
    )
  ) |> dplyr::bind_rows()
```



Note that the inner sum on  $t$  gives  $s - 1$  pairs, so that the total number of pairs is

$$\sum_{s=2}^T (s-1) = 1 + 2 + 3 + \dots + (T-1) = \frac{T(T-1)}{2}$$

or half of all  $T^2$  index pairs  $(t, s)$  less the diagonal  $(s, s)$ . From figure (B) we see that it sums the lower diagonal of all index pair values, less the diagonal.

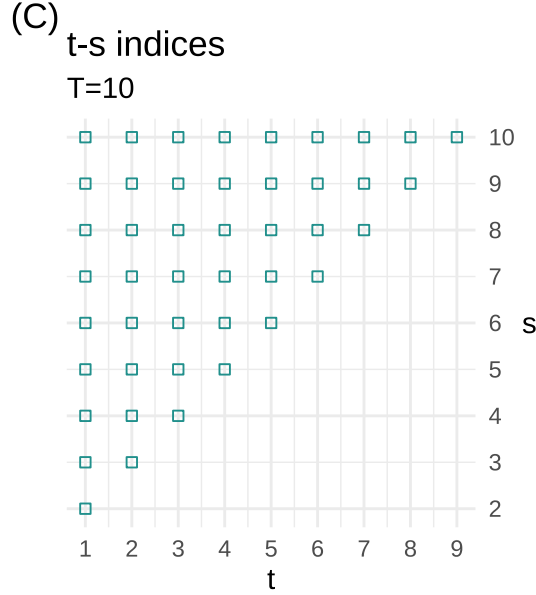
This sum also counts the number of unique pairs  $(s, t)$  where  $2 \leq s \leq T$  and  $1 \leq t < s$  which is equivalent to counting the number of pairs where  $1 \leq t < s \leq T$

**term 4:**

The fourth term is  $-\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} a_t b_{t+k}$  and again we will use the change of variables  $s = t + k$ .

With the change of variable, and with  $s$  now the second index, the inner sum goes from  $s = t + 1$  to  $s = T$  and the outer sum goes from  $t = 1$  to  $t = T - 1$ . The  $(t, s)$  indices used to sum  $-\frac{1}{T^2} \sum_{t=1}^{T-1} \sum_{s=t+1}^T a_t b_s$  are shown in figure (B) below (for  $T = 10$ ).

In fact the  $a_t b_{t+k}$  terms are the transpose of the terms  $a_{t+k} b_t$  of figure (B) as can be seen in figure (C).



So per terms 3 & 4, each cross-product ( $a_i b_j$ ,  $i \neq j$ ) appears exactly once with a negative sign, for a total of  $T^2 - T$  terms (since each of terms 3 & 4 has  $\frac{T(T-1)}{2}$  terms).

So we are missing a set of  $T$  diagonal terms from the total  $T^2$  terms.

but the sum of the first two terms is  $\frac{T-1}{T^2} \sum_{t=1}^T a_t b_t$ , which provides the missing diagonal (i.e.  $-\frac{1}{T^2} \sum_{t=1}^T a_t b_t$ ), so we have:

$$\frac{T-1}{T^2} \sum_{t=1}^T a_t b_t - \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1, t \neq s}^T a_t b_s = \quad (9)$$

$$\frac{1}{T} \sum_{t=1}^T a_t b_t - \frac{1}{T^2} \sum_{s=1}^T \sum_{t=1}^T a_t b_s = \quad (10)$$

$$\frac{1}{T} \sum_{t=1}^T a_t b_t - \frac{1}{T} \sum_{s=1}^T b_s \times \frac{1}{T} \sum_{t=1}^T a_t = \quad (11)$$

$$\frac{1}{T} \sum_{t=1}^T a_t b_t - \bar{b} \bar{a} \quad (12)$$

compute example for covariance:

```
set.seed(8740); T = 20
a <- 1 + rnorm(n = T)
b <- a + 4 + rnorm(n = T, sd = 2)

# calculate using difference equation
cov_diff <-
  1:(T-1) |> purrr::map_vec(
    \(k){ ((dplyr::lead(a,k) - a) * (dplyr::lead(b,k) - b)) |> sum(na.rm=TRUE) })
  ) |> sum(na.rm=TRUE) / (T*T)

# summarize results
tibble::tibble(x = a, y = b) |>
  dplyr::mutate(x = x - mean(x), y = y - mean(y), prod = x*y) |>
  dplyr::summarize(pop_cov = mean(prod) ) |>
  tibble::add_column(cov_diff = cov_diff) |>
  gt::gt() |>
  gt::tab_header(
    title = "Covariance Calculations"
    , subtitle = stringr::str_glue("T={T}") |>
  gt::cols_width( everything() ~ gt::px(150) ) |>
  gt::cols_label(
    pop_cov = gt::md("using product formula")
    , cov_diff = "using diff formula"
  ) |>
  gt::tab_options(table.width = gt::pct(50), table.align = "center") |>
  gtExtras::gt_theme_espn() #|> gt::as_latex()
```

## Covariance Calculations

T=20

| using product<br>formula | using diff formula |
|--------------------------|--------------------|
| 0.9179426                | 0.9179426          |

### compute example for covariance:

The difference formula for (population) covariance is

$$\frac{1}{T^2} \sum_{k=1}^{T-1} \sum_{t=1}^{T-k} (a_{t+k} - a_t) (b_{t+k} - b_t)$$

And the same expression holds for variance calculations: the population variance is calculated as the mean of the square of centered values, can also be calculated using all the differences between values.

```
a_var <- (a-mean(a))^2 |> mean()

# calculate using difference equation
var_diff <-
  1:(T-1) |> purrr::map_vec(
    \(k){ ((dplyr::lead(a,k) - a)^2) |> sum(na.rm=TRUE) })
  ) |> sum(na.rm=TRUE) / (T*T)

# summarize results
tibble::tibble("using product formula" = a_var, "using difference formula" = var_diff) |>
  gt::gt() |>
  gt::tab_header(
    title = "Variance Calculations"
    , subtitle = stringr::str_glue("with variable a | T={T}")
  ) |>
  gt::tab_options(table.width = gt::pct(50), table.align = "center") |>
  gtExtras::gt_theme_espn()
```

#### Variance Calculations with variable a | T=20

| using product formula | using difference formula |
|-----------------------|--------------------------|
| 0.6150733             | 0.6150733                |